

M A P W I S E

EVERY PIXEL OF THE EARTH, CLASSIFIED

A U-Net that reads satellite imagery and labels every pixel as urban, agriculture, forest, water, rangeland or barren — then reports how much ground each one covers.

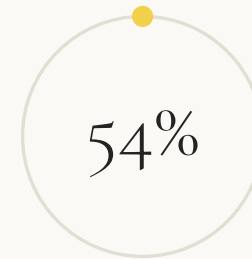
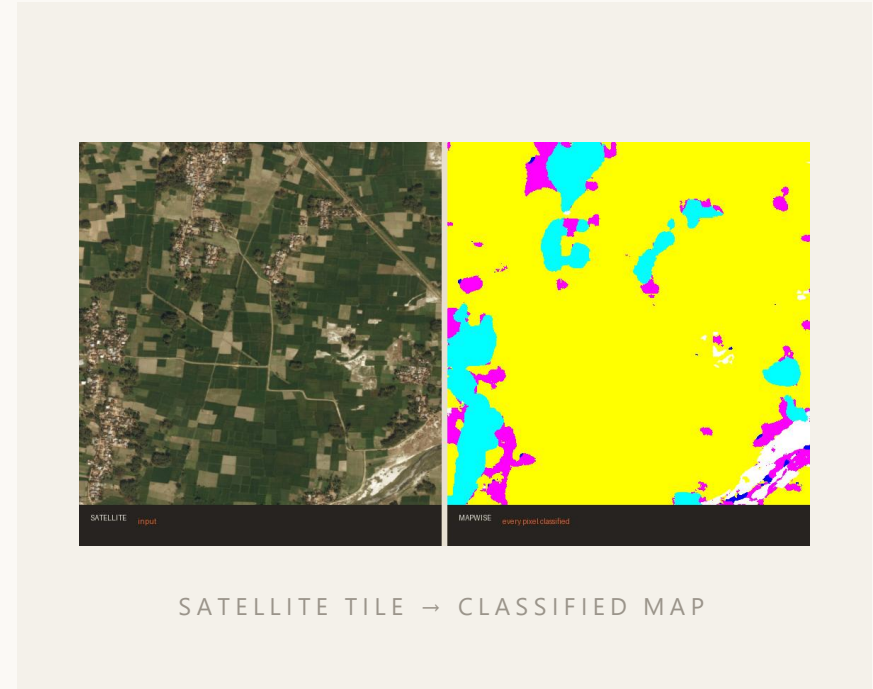
PRIYANSHU SINGH · AI / ML ENGINEER

THE IDEA

Satellites see everything. *Nobody can look at it all.*

Terabytes of imagery arrive every day. The questions asked of it are simple — how much forest was lost, where did the city expand, which fields are planted — but answering them by eye does not scale.

MapWise answers them per pixel, automatically, and returns the percentages rather than just a picture.



MEAN IoU



PIXEL ACCURACY

ARCHITECTURE

How it works

A U-Net: a pretrained encoder reads the image, a decoder written from scratch rebuilds it as labels.

01

ENCODE

A ResNet-34 compresses the tile into a feature pyramid, from fine texture to coarse context.

02

DECODE

Four upsampling blocks rebuild full resolution, each re-joining the matching encoder detail.

03

CLASSIFY

A 1x1 convolution turns every pixel's features into seven class scores.

04

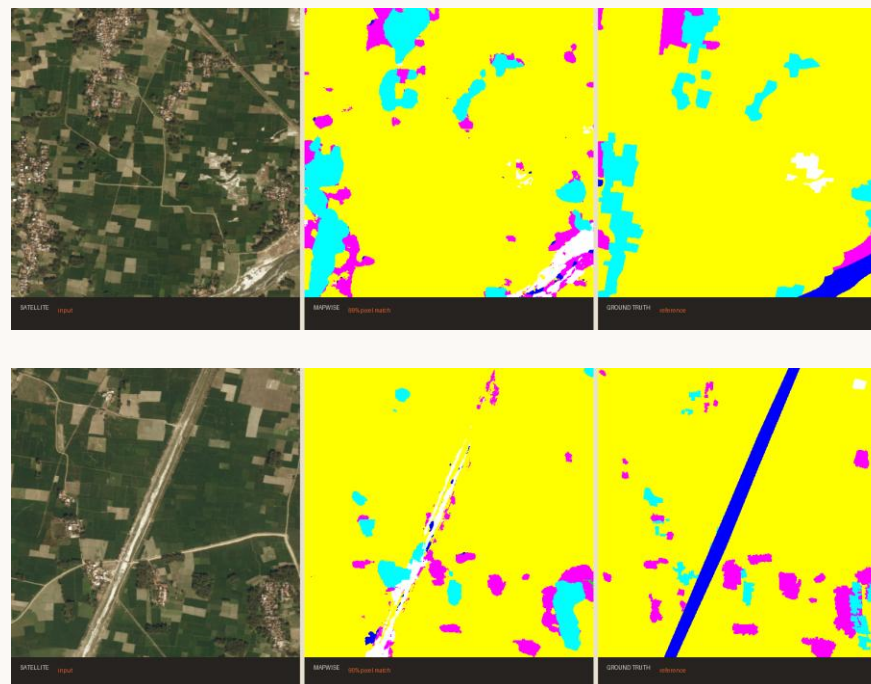
MEASURE

Pixel counts become ground-cover percentages — the number a client actually uses.

Skip connections are why the edges stay sharp.

RESULTS IN THE FIELD

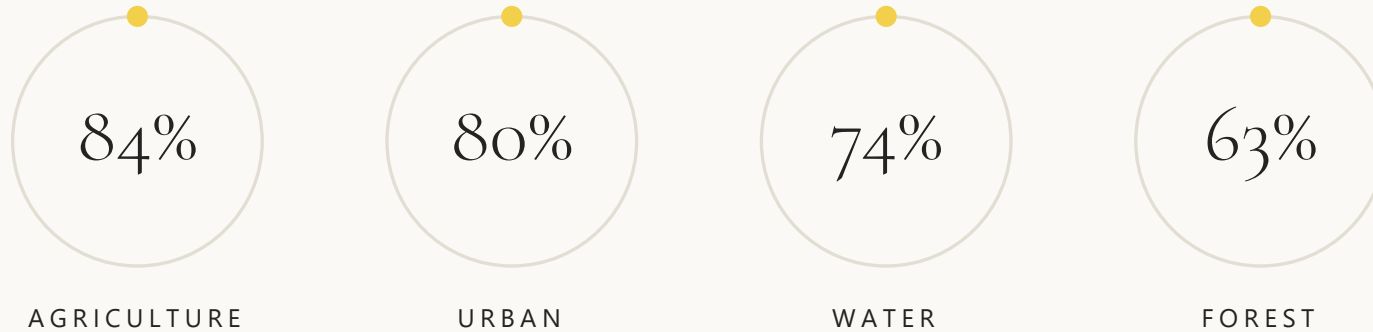
Satellite in, map out



Left: the satellite tile. Centre: MapWise. Right: the reference map a human analyst drew.

ACCURACY, CLASS BY CLASS

What it gets right — *and what it doesn't*



The rare classes are hard.

DeepGlobe is heavily imbalanced. Weighted Dice loss lifts rangeland (34%) and barren (45%) into usefulness, but 'unknown' is a few dozen pixels a tile and stays at zero.

So the numbers are published per class, not averaged away.

Mean IoU 54% · pixel accuracy 84% · 24 epochs in 41 minutes on one T4

PROCESS

How it's built

01



DATA

2448px tiles, random 512px crops, flips and rotations — aerial imagery has no 'up'.

02



ARCHITECTURE

ResNet-34 encoder, U-Net decoder written from scratch.

03



TRAINING

Dice + weighted cross-entropy, mixed precision, cosine schedule.

04



DEPLOY

ONNX export, int8 quantisation, inference inside the browser.

WHO IT'S FOR

The mask is the demo.
The number is the product.

AGRICULTURE

Planted area per field, season over season,
without anyone walking the land.

URBAN PLANNING

Where the built-up edge moved,
measured rather than estimated.

ENVIRONMENT & RISK

Deforestation, water extent and wildfire
fuel — inputs insurers price on.

M A P W I S E

See the ground as data.

Try it on your own satellite tile. The model runs inside your browser,
so the image never leaves your device.

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github.com/Priyanshu-Singh-git/mapwise



SCAN TO TRY IT LIVE

priyanshu-singh-git.github.io/mapwise